MATH 201 Your name: _____

Learning target 2.1, version 1

Compute derivatives using basic function rules, the constant multiple rule, and the sum rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(x) = 2\cos(x) + 4e^x$

2. $h(t) = \sqrt[3]{t^4} + \dfrac{4}{t^4}$ (Don't use the quotient rule or the chain rule.)

3. $g(w) = 6w^3 - 2w^2 - 9w - 2$

4. $C(t) = 64(e^7 + 7^e) - \dfrac{\sqrt[27]{193262}}{16 - \pi^2} \cdot \sin(4)$ (Hint: Look carefully!)

MATH 201 Your name: _____

Learning target 2.1, version 2

Compute derivatives using basic function rules, the constant multiple rule, and the sum rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $g(t) = \sqrt[4]{t^3} + \dfrac{7}{t^4}$ (Don't use the quotient rule or the chain rule.)

2. $f(x) = -x^5 - 4x^4 + 5x - 4$

3. $h(y) = -4\ln(y) + 3\sin(y)$

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Learning target 2.1, version 3

Compute derivatives using basic function rules, the constant multiple rule, and the sum rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $f(y) = -2e^y - 4\sin(y)$

2. $h(x) = 9x^5 - x^4 - x^3 - 6$

3. $g(t) = \sqrt[3]{t^7} + \dfrac{2}{t^2}$ (Don't use the quotient rule or the chain rule.)

MATH 201 Your name: _____

Learning target 2.1, version 4

Compute derivatives using basic function rules, the constant multiple rule, and the sum rule.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (constant multiple, sum/difference, etc.) you are using in your work.

1. $g(t) = -5t^4 - 9t^3 - 6t^2 - 6$

2. $f(w) = \sqrt[3]{w^5} + \dfrac{4}{w^5}$ (Don't use the quotient rule or the chain rule.)

3. $h(y) = 2e^y + 2\sin(y)$